화학수학

열여덟 번째 수업

확률

Point Group (점군)

Formed by a set of point symmetry operators

belonging to a symmetrical object

Representation of a Point Group

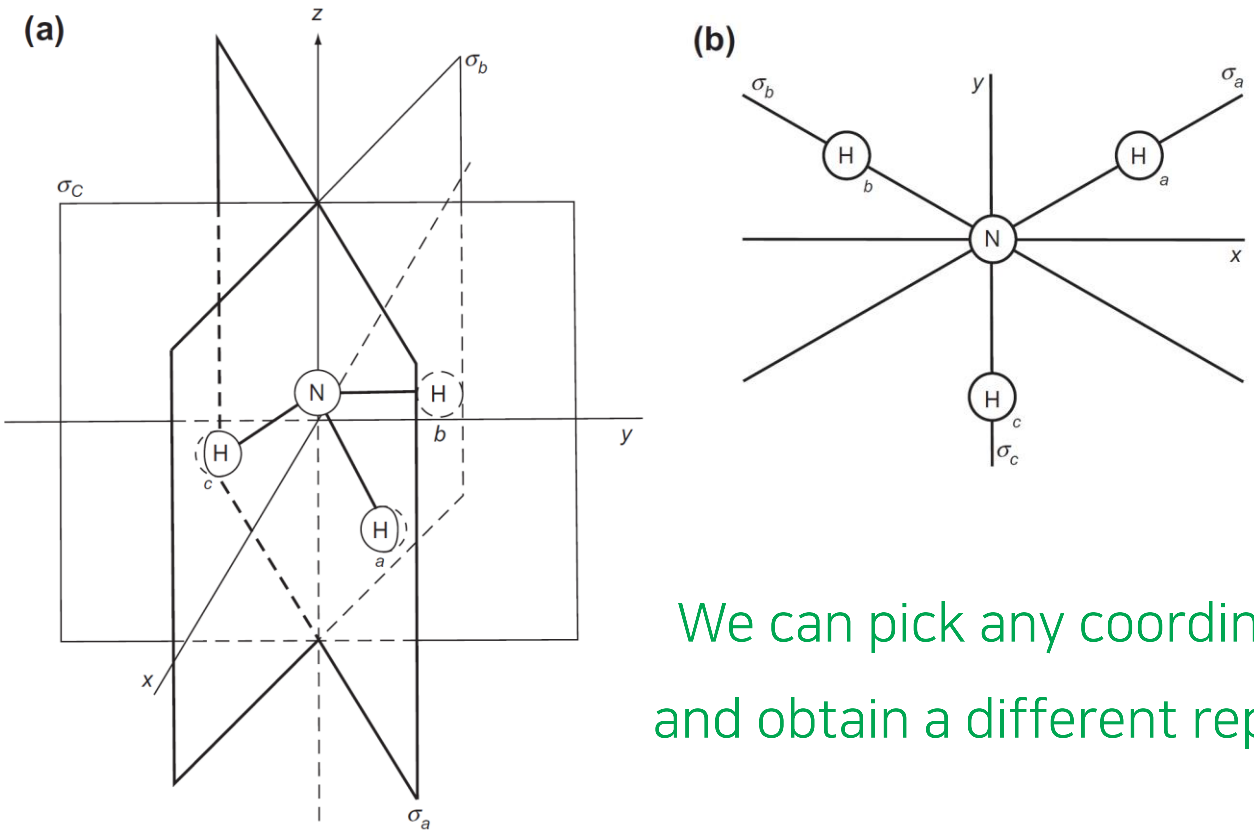
Symmetry operator ↔ Matrix

$$\hat{A}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\mathbf{A}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

Arbitrary Nature of Representation



We can pick any coordinate system and obtain a different representation

(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.4)

Similarity Transformation (닮음 변환)

- Two different representations $\mathbf{P}$ and $\mathbf{Q}$ in different coordinates can be linked by

$$\mathbf{P} = \mathbf{X}^{-1}\mathbf{Q}\mathbf{X}$$

  when $\mathbf{X}$ is some square matrix

- Since $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$,

$$\mathrm{tr}(\mathbf{P}) = \mathrm{tr}(\mathbf{X}^{-1}\mathbf{Q}\mathbf{X}) = \mathrm{tr}(\mathbf{X}\mathbf{X}^{-1}\mathbf{Q}) = \mathrm{tr}(\mathbf{Q})$$

Character (지표)

- Character of a symmetry operator = trace of the corresponding matrix

$$\hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A}$$

Character of $\hat{C}_3 = \mathrm{tr}\,\mathbf{A} = (-1/2) + (-1/2) + 1 = 0$

Reducible Representation (가약 표현)

All matrices are block-diagonal, or can be put into block-diagonal form by a similarity transformation

$$\hat{E} \leftrightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{E}, \qquad \hat{C}_3 \leftrightarrow \begin{bmatrix} -1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{A},$$

$$\hat{C}_3^2 \leftrightarrow \begin{bmatrix} -1/2 & \sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{B}, \qquad \hat{\sigma}_a \leftrightarrow \begin{bmatrix} 1/2 & \sqrt{3}/2 & 0 \\ \sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{C},$$

$$\hat{\sigma}_b \leftrightarrow \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ -\sqrt{3}/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{D}, \qquad \hat{\sigma}_c \leftrightarrow \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \mathbf{F}$$

Irreducible Representation (기약 표현)

Cannot be made block-diagonal

by any similarity transformation

- Reduction of a representation (표현의 약분): to make the matrices in the representation block-diagonal by some similarity transformation
- Often abbreviated as irrep

Characters of Irreps

Class: E $2C_3$ $3\sigma_v$

Irrep	$\hat{E}$	$\hat{C}_3$	$\hat{C}_3^2$	$\hat{\sigma}_a$	$\hat{\sigma}_b$	$\hat{\sigma}_c$
2 by 2	2	-1	-1	0	0	0
1 by 1	1	1	1	1	1	1

Class is constituted of symmetry operations that are "physically identical"

"Full" Character Table of C_{3v}

C_{3v}	E	$2C_3$	$3\sigma_v$	Basis components		
A_1	1	1	1	z		x^2+y^2, z^2
A_2	1	1	-1		R_z	
E	2	-1	0	(x, y)	(R_x, R_y)	(x^2-y^2, xy) (xz, yz)

Application of Character Table

- To make a chemical bond, we need

$$\int \psi_i^* \psi_j \, \mathrm{d}\tau \neq 0$$

- According to the group theory, the integral may be nonzero only if ψ_i and ψ_j belong to the same irrep of the molecular point group

오늘의 수업

- Probability distributions
  - Discrete probability distributions
  - Continuous probability distributions

- Examples
  - Binomial distribution
  - Uniform distribution
  - Gaussian distribution

15. Probability, Statistics, and Experimental Errors

(15장 확률, 통계, 실험오차)

Probability Theory

study of random events

- Classical systems: unknown information
- Quantum systems: intrinsic randomness

- Population: large set of objects or numbers
  - Classical systems: imaginary set of infinitely many repetitions of a given measurement
  - Quantum systems: imaginary set of infinitely many replicas of the physical system

Not in Mortimer

Probability Theory in Chemistry

When you have a set of different events

- Quantum mechanics: different quantum states
- Kinetic theory: different molecular speeds
- Statistical mechanics: different systems
- Error analysis: different measurements

Probability Distribution (확률분포)

mathematical function that gives

the probabilities of occurrence of

different possible outcomes for an experiment

- Discrete probability distribution (이산 확률분포): possible outcomes have discrete values
- Continuous probability distribution (연속 확률분포): possible outcomes have continuous values

Discrete Probability Distribution

- Discrete set of values: $x_1, x_2, x_3, \cdots$

Probability of x_i = $p_i \geq 0$

- *e.g.* fair dice: $p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = 1/6$

- Total probability

$$p_{\text{tot}} = \sum_{i=1}^{\infty} p_i$$

Mean Value and Normalization

- Mean value (평균값)

$$\langle x \rangle = \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i x_i$$

- Normalization (정규화): $p_{\text{tot}} = \sum_{i=1}^{\infty} p_i = 1$
- Mean value with a normalized distribution

$$\langle x \rangle = \sum_{i=1}^{\infty} p_i x_i$$

Example 15.2

(1) Find the probability of each of the outcomes of tossing two unloaded dice and (2) find the mean value.

The outcomes range from 2 to 12.

The total number of possible ways is 6 × 6 = 36.

$$\langle x \rangle = \sum_{i=2}^{12} p_i x_i$$

$$\langle x \rangle = 7$$

Total (n)	Ways to achieve	Number of ways	Normalized probability $= p_n$
2	1 and 1	1	$1/36 = 0.02778$
3	1 and 2, 2 and 1	2	$2/36 = 0.05556$
4	1 and 3, 2 and 2, 3 and 1	3	$3/36 = 0.08333$
5	1 and 4, 2 and 3, 3 and 2, 4 and 1	4	$4/36 = 0.11111$
6	1 and 5, 2 and 4, 3 and 3, 4 and 2, 5 and 1	5	$5/36 = 0.13889$
7	1 and 6, 2 and 5, 3 and 4, 4 and 3, 5 and 2, 6 and 1	6	$6/36 = 0.16667$
8	2 and 6, 3 and 5, 4 and 4, 5 and 3, 2 and 6	5	$5/36 = 0.13889$
9	3 and 6, 4 and 5, 5 and 4, 6 and 3	4	$4/36 = 0.11111$
10	4 and 6, 5 and 5, 6 and 4	3	$3/36 = 0.08333$
11	5 and 6, 6 and 5	2	$2/36 = 0.05556$
12	6 and 6	1	$1/36 = 0.02778$

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Expected Value (기댓값)

For a function $g(x_i)$ that depends on x_i,

the expected value is

$$\langle g \rangle = \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i g(x_i)$$

e.g. dice game with $g(1) = ₩30{,}000$

$$\langle g \rangle = \frac{1}{6} \times (₩30{,}000) + \frac{1}{6} \times 0 + \cdots + \frac{1}{6} \times 0 = ₩5{,}000$$

Variance (분산)

$$\sigma_x^2 = \langle (x - \langle x \rangle)^2 \rangle$$

$$\sigma_x^2 = \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i (x_i - \langle x \rangle)^2 = \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i \left(x_i^2 - 2\langle x \rangle x_i + \langle x \rangle^2 \right)$$

$$= \frac{1}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i x_i^2 - \frac{2\langle x \rangle}{p_{\text{tot}}} \sum_{i=1}^{\infty} p_i x_i + \frac{\langle x \rangle^2}{\cancel{p_{\text{tot}}}} \cancel{\sum_{i=1}^{\infty} p_i}$$

$$= \langle x^2 \rangle - 2\langle x \rangle^2 + \langle x \rangle^2 = \langle x^2 \rangle - \langle x \rangle^2$$

Example 15.3

Find the variance for the distribution of dice throws in the previous example.

We know: $\langle x \rangle = 7$

Noting that $\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2$,

$$\langle x^2 \rangle = \frac{329}{6}$$

Hence,

$$\sigma_x^2 = \frac{329}{6} - 7^2 = \frac{35}{6}$$

Total (n)	Ways to achieve	Number of ways	Normalized probability $= p_n$
2	1 and 1	1	$1/36 = 0.02778$
3	1 and 2, 2 and 1	2	$2/36 = 0.05556$
4	1 and 3, 2 and 2, 3 and 1	3	$3/36 = 0.08333$
5	1 and 4, 2 and 3, 3 and 2, 4 and 1	4	$4/36 = 0.11111$
6	1 and 5, 2 and 4, 3 and 3, 4 and 2, 5 and 1	5	$5/36 = 0.13889$
7	1 and 6, 2 and 5, 3 and 4, 4 and 3, 5 and 2, 6 and 1	6	$6/36 = 0.16667$
8	2 and 6, 3 and 5, 4 and 4, 5 and 3, 2 and 6	5	$5/36 = 0.13889$
9	3 and 6, 4 and 5, 5 and 4, 6 and 3	4	$4/36 = 0.11111$
10	4 and 6, 5 and 5, 6 and 4	3	$3/36 = 0.08333$
11	5 and 6, 6 and 5	2	$2/36 = 0.05556$
12	6 and 6	1	$1/36 = 0.02778$

Standard Deviation (표준편차)

$$\sigma_x = \sqrt{\sigma_x^2} = [\langle x^2 \rangle - \langle x \rangle^2]^{1/2}$$

Continuous Probability Distributions

- Continuous value: x
- Continuous probability distribution (probability density): $f(x)$

  $f(x')\mathrm{d}x$ = Probability of values of $x \in [x', x' + dx]$

- Probability that $c < x < d$ = $\int_c^d f(x)\,\mathrm{d}x$

Total Probability

$$\int_{x_{\min}}^{x_{\max}} f(x)\,\mathrm{d}x$$

- Normalization

$$\int_{x_{\min}}^{x_{\max}} f(x)\,\mathrm{d}x = 1$$

Mean Value

$$\langle x \rangle = \frac{\int_{x_{\min}}^{x_{\max}} x f(x)\,\mathrm{d}x}{\int_{x_{\min}}^{x_{\max}} f(x)\,\mathrm{d}x}$$

- Mean value with a normalized distribution

$$\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x)\,\mathrm{d}x$$

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Expected Value

- For a function $g(x)$, the expected value is

$$\langle g \rangle = \frac{\int_{x_{\min}}^{x_{\max}} f(x) g(x)\, \mathrm{d}x}{\int_{x_{\min}}^{x_{\max}} f(x)\, \mathrm{d}x}$$

- e.g. Mean-square value

$$\langle x^2 \rangle = \frac{\int_{x_{\min}}^{x_{\max}} x^2 f(x)\, \mathrm{d}x}{\int_{x_{\min}}^{x_{\max}} f(x)\, \mathrm{d}x}$$

Variance and Standard Deviation

- Variance

$$\sigma_x^2 = \langle (x - \langle x \rangle)^2 \rangle = \frac{\int_{x_{\min}}^{x_{\max}} (x - \langle x \rangle)^2 f(x)\,\mathrm{d}x}{\int_{x_{\min}}^{x_{\max}} f(x)\,\mathrm{d}x}$$

- Standard deviation

$$\sigma_x = \sqrt{\sigma_x^2}$$

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Summary

	Discrete	Continuous
Variable	$x_1, x_2, \cdots, x_n$	x: real in $[x_{\min}, x_{\max}]$
Probability distribution	p_i	$f(x)\,\mathrm{d}x$
Normalization	$\sum_{i=1}^{n} p_i = 1$	$\int_{x_{\min}}^{x_{\max}} f(x)\,\mathrm{d}x = 1$
Mean value (normalized dist.)	$\langle x \rangle = \sum_{i=1}^{n} x_i p_i$	$\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x)\,\mathrm{d}x$
Expected value (normalized dist.)	$\langle g \rangle = \sum_{i=1}^{n} g(x_i) p_i$	$\langle g \rangle = \int_{x_{\min}}^{x_{\max}} g(x) f(x)\,\mathrm{d}x$

Examples

- Discrete probability distribution
  - Binomial probability distribution
- Continuous probability distribution
  - Uniform probability distribution
  - Gaussian probability distribution

Binomial Probability Distribution

(이항확률분포)

Toss an unbiased coin n times, and

find the probability that "heads" will come up m times

Probability Calculation

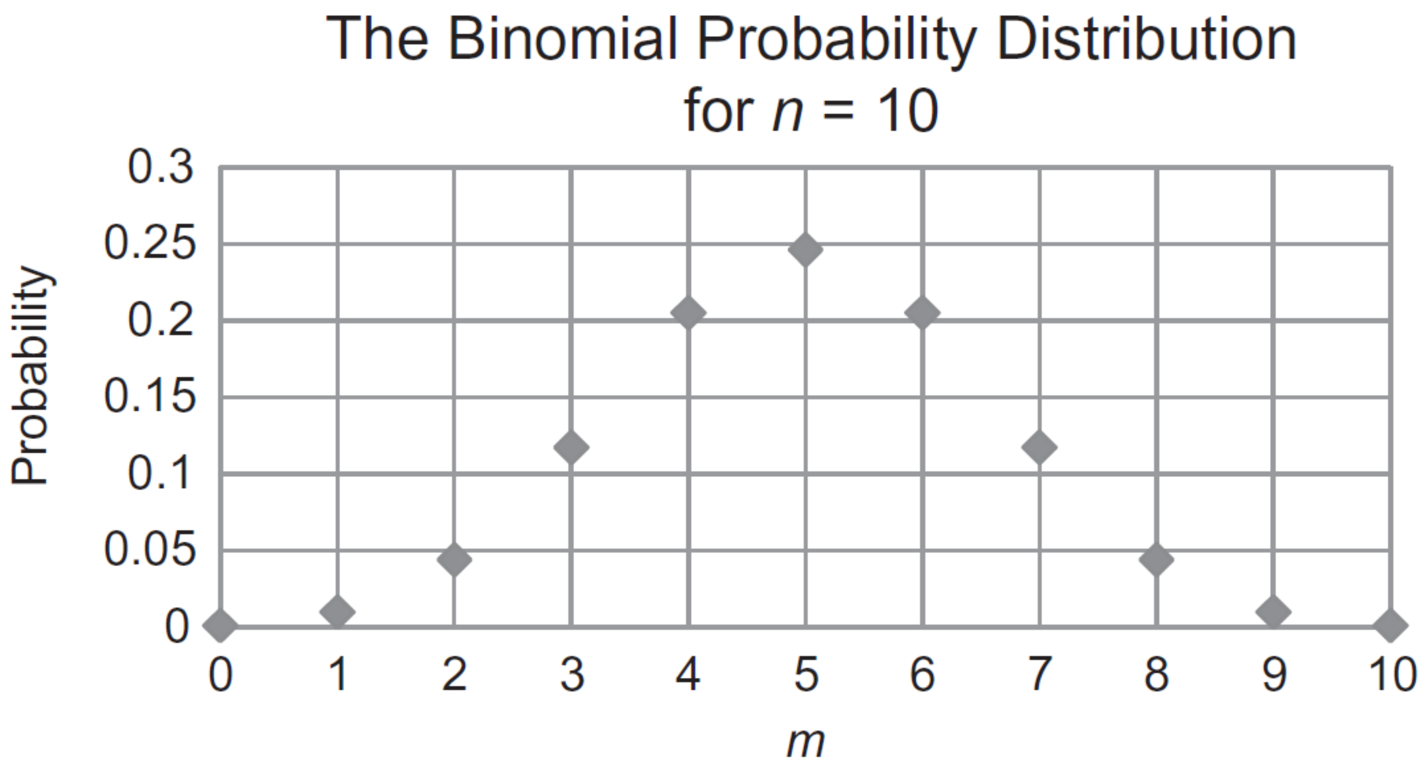
- Number of choosing m coins among n coins

$$ {}_{n}\mathrm{C}_{m} = \binom{n}{m} = \frac{n!}{m!\,(n-m)!} $$

- Probability for a "head" = probability for a "tail" = 1/2
- Final probability

$$ P(n,m) = \frac{n!}{m!\,(n-m)!}\left(\frac{1}{2}\right)^{m}\left(\frac{1}{2}\right)^{n-m} = \frac{n!}{m!\,(n-m)!}\left(\frac{1}{2}\right)^{n} $$

Plotting the Distribution



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 15.1)

Example 15.4

Calculate the probability that "heads" will come up 50 times if an biased coin is tossed 100 times.

$$p(n,m) = \frac{n!}{m!\,(n-m)!}\left(\frac{1}{2}\right)^{n}$$

Here, $n = 100$ and $m = 50$, and

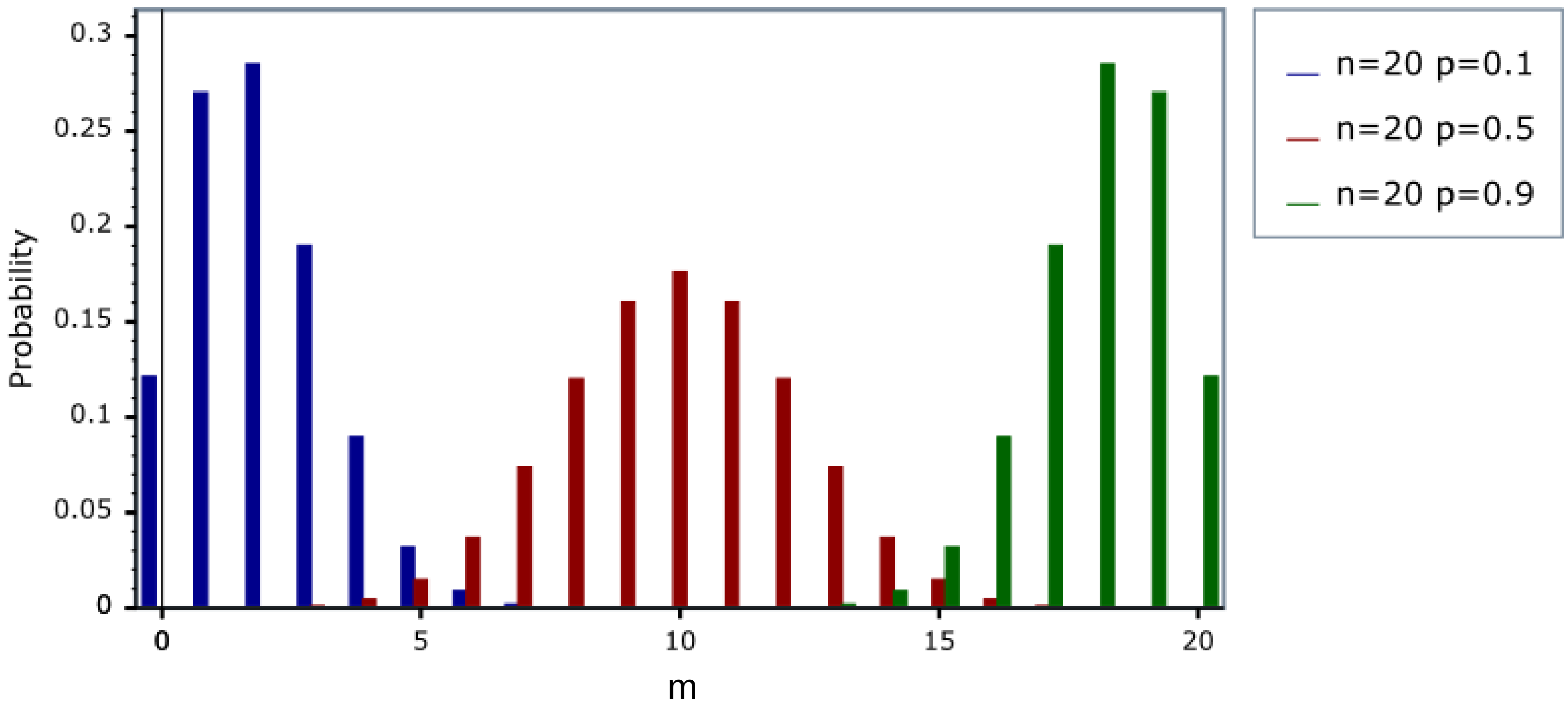
$$p(100{,}50) = \frac{100!}{50!\,50!}\left(\frac{1}{2}\right)^{100} = 0.07959\cdots$$

With Biased Coins

- p = probability that "heads" will occur
- $(1-p)$ = probability that "tails" will occur

$$P(n,m) = \frac{n!}{m!\,(n-m)!}p^m(1-p)^{n-m}$$

Binomial Probability Distributions



(Image: https://www.boost.org/doc/libs/1_42_0/libs/math/doc/sf_and_dist/graphs/binomial_pdf_2.png)

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Mean Value and Variance

- Mean value

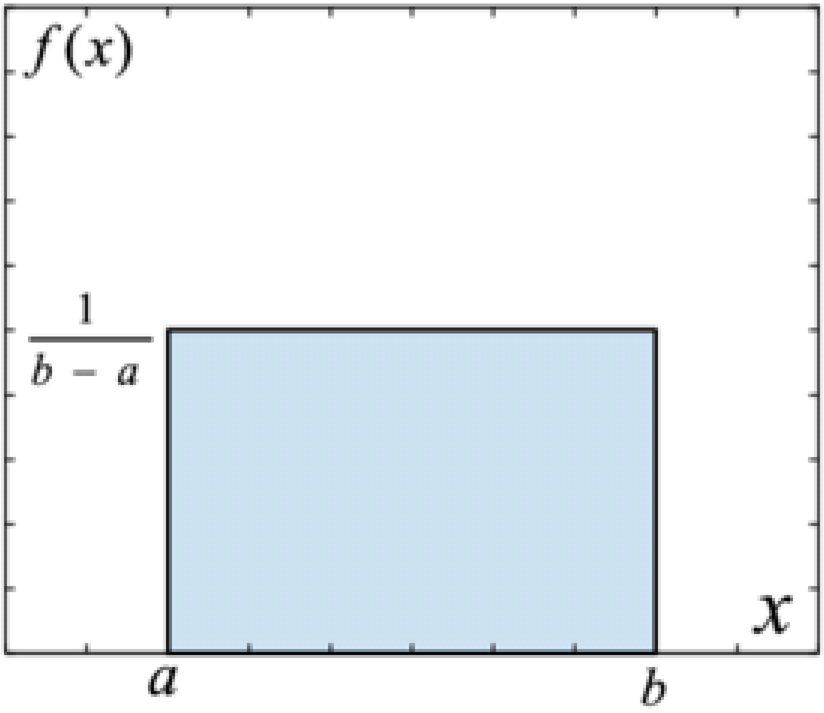
$$\langle x \rangle = \sum_{i=1}^{\infty} p_i x_i = \sum_{m=0}^{n} \frac{n!}{m!\,(n-m)!} p^m (1-p)^{n-m} \times m$$

$$\langle x \rangle = np$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = np(1-p)$$

Uniform Probability Distribution



(Image: https://www.researchgate.net/figure/Probability-distribution-function-a-uniform-distribution-of-the-excitation-frequency_fig5_339429775)

Mean Value and Mean-Square Value

- Mean value

$$\langle x \rangle = \int_a^b x f(x)\,\mathrm{d}x = \int_a^b \frac{x}{b-a}\,\mathrm{d}x = \frac{1}{b-a}\left[\frac{1}{2}x^2\right]_a^b = \frac{a+b}{2}$$

- Mean-square value

$$\langle x^2 \rangle = \int_a^b x^2 f(x)\,\mathrm{d}x = \int_a^b \frac{x^2}{b-a}\,\mathrm{d}x = \frac{1}{b-a}\left[\frac{1}{3}x^3\right]_a^b$$

$$= \frac{a^2+ab+b^2}{3}$$

Variance

$$\langle x \rangle = \frac{a+b}{2}; \qquad \langle x^2 \rangle = \frac{a^2+ab+b^2}{3}$$

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2$$

$$= \frac{a^2-2ab+b^2}{12} = \frac{(a-b)^2}{12}$$

Example 15.5

Find the mean, the variance, and the standard deviation for a uniform probability distribution with a = 0 and b = 10. Find the probability that x lies between $\langle x\rangle - \sigma_x$ and $\langle x\rangle + \sigma_x$.

$$\langle x\rangle = \frac{a+b}{2} = \frac{0+10}{2} = 5$$

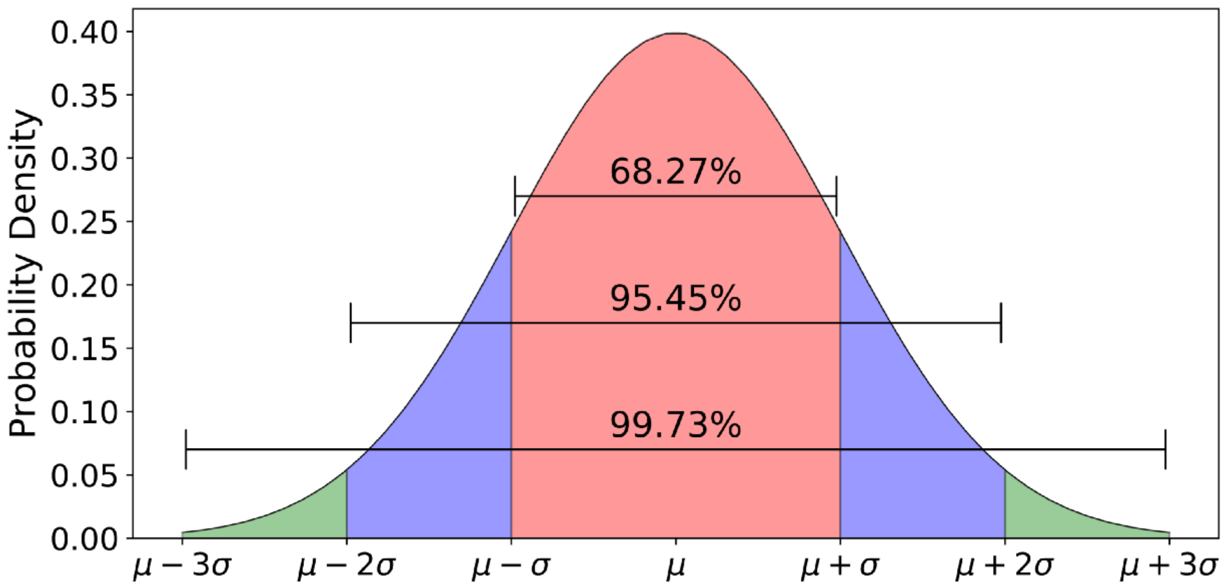
$$\sigma_x^2 = \frac{(a-b)^2}{12} = \frac{(0+10)^2}{12} = \frac{25}{3}$$

$$\sigma_x = \sqrt{\sigma_x^2} = \sqrt{\frac{25}{3}} = \frac{5}{\sqrt{3}}$$

$$\text{Probability} = \int_{\langle x\rangle-\sigma_x}^{\langle x\rangle+\sigma_x} f(x)\,\mathrm{d}x = \int_{5-5/\sqrt{3}}^{5+5/\sqrt{3}} \frac{1}{10-0}\,\mathrm{d}x = \frac{1}{\sqrt{3}}$$

Gaussian Probability Distribution

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$



(Image: https://towardsdatascience.com/understanding-the-68-95-99-7-rule-for-a-normal-distribution-b7b7cbf760c2)

Example 15.6

Show that the Gaussian distribution satisfies the normalization condition.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

The normalization condition states

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$$

Letting $t = (x-\mu)/\sqrt{2\sigma^2}$,

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} e^{-t^2} \left(\sqrt{2\sigma^2}\,\mathrm{d}t\right) = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-t^2}\,\mathrm{d}t = 1$$

Hence, the Gaussian distribution satisfies the normalization condition.

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Mean Value and Variance

- Mean value

$$\langle x \rangle = \int_{-\infty}^{\infty} x f(x)\,\mathrm{d}x = \mu$$

- Variance

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2 = \sigma^2$$

Error Function (오차함수)

$$\mathrm{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2}\,\mathrm{d}t$$

(Fraction between $\mu - \alpha$ and $\mu + \alpha$)

$$= \int_{\mu-\alpha}^{\mu+\alpha} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}\,\mathrm{d}x = \int_{-\alpha/\sqrt{2\sigma^2}}^{\alpha/\sqrt{2\sigma^2}} \frac{1}{\sqrt{\pi}} e^{-t^2}\,\mathrm{d}t$$

$$= \frac{2}{\sqrt{\pi}} \int_0^{\alpha/\sqrt{2\sigma^2}} e^{-t^2}\,\mathrm{d}t = \mathrm{erf}\left(\frac{\alpha}{\sqrt{2\sigma^2}}\right)$$

Appendix G

Values of the Error Function

x		0	1	2	3	4	5	6	7	8	9
0.0	0.0	000	113	226	338	451	564	676	789	901	013*
0.1	0.1	125	236	348	459	569	680	790	900	009*	118*
0.2	0.2	227	335	443	550	657	763	869	974	079*	183*
0.3	0.3	286	389	491	593	694	794	893	992	090*	187*
0.4	0.4	284	380	475	569	662	755	847	937	027*	117*
0.5	0.5	205	292	379	465	549	633	716	798	879	959
0.6	0.6	039	117	194	270	346	420	494	566	638	708
0.7		778	847	914	981	047*	112*	175*	238*	300*	361*
0.8	0.7	421	480	538	595	651	707	761	814	867	918
0.9		969	019*	068*	116*	163*	209*	254*	299*	342*	385*
1.0	0.8	427	468	508	548	586	624	661	698	733	768
1.1		802	835	868	900	931	961	991	020*	048*	076*
1.2	0.9	103	130	155	181	205	229	252	275	297	319
1.3		340	361	381	400	419	438	456	473	490	507
1.4	0.95	23	39	54	69	83	97	11*	24*	37*	49*
1.5	0.96	61	73	84	95	06*	16*	26*	36*	45*	55*
1.6	0.97	63	72	80	88	96	04*	11*	18*	25*	32*
1.7	0.98	38	44	50	56	61	67	72	77	82	86
1.8		91	95	99	03*	07*	11*	15*	18*	22*	25*
1.9	0.99	28	31	34	37	39	42	44	47	49	51
2.0	0.995	32	52	72	91	09*	26*	42*	58*	73*	88*
2.1	0.997	02	15	28	41	53	64	75	85	95	05*
2.2	0.998	14	22	31	39	46	54	61	67	74	80
2.3		86	91	97	02*	06*	11*	15*	20*	24*	28*
2.4	0.999	31	35	38	41	44	47	50	52	55	57
2.5		59	61	63	65	67	69	71	72	74	75
2.6		76	78	79	80	81	82	83	84	85	86
2.7		87	87	88	89	89	90	91	91	92	92
2.8	0.9999	25	29	33	37	41	44	48	51	54	56
2.9		59	61	64	66	68	70	72	73	75	77

* From Eugene Jahnke and Fritz Emde, *Tables of Functions*, Dover Publications, New York, 1945, p. 24.

Example 15.7

Find the probability that x lies between $\mu - 0.5\sigma$ and $\mu + 0.5\sigma$ for a Gaussian distribution.

$$\text{(Fraction between } \mu - \alpha \text{ and } \mu + \alpha) = \operatorname{erf}\left(\frac{\alpha}{\sqrt{2\sigma^2}}\right)$$

Thus,

$$\text{Probability} = \operatorname{erf}\left(\frac{0.5\sigma}{\sqrt{2\sigma^2}}\right) = \operatorname{erf}\left(\frac{1}{2\sqrt{2}}\right) = \operatorname{erf}(0.35) = 0.38$$

Central Limit Theorem (중심극한정리)

For random variables $x_1, x_2, \cdots, x_n$,

$$y = x_1 + x_2 + \cdots + x_n$$

is governed by a probability distribution that approaches a Gaussian distribution for large n

- If experimental errors arise from multiples sources, they should be at least approximately described by a Gaussian distribution